

Paolo L. Sinopoli

Education

Massachusetts Institute of Technology

September 2026 - Present

Biology Ph.D. Student

National Science Foundation Graduate Fellow

The Ohio State University

August 2022 – May 2026

Bachelor of Science in Molecular Genetics (*with Honors Research Distinction*)

GPA: 4.00/4.00

Provost Scholarship

Research Positions

Massachusetts Institute of Technology, Cambridge, MA

June 2026 – September 2026

Research Intern (Advisor: Dr. Anders Sejr Hansen)

Project: Towards super-resolution imaging of Polycomb bodies and Polycomb-mediated 3D genome interactions

The Ohio State University, Columbus, OH

October 2022 – May 2026

Undergraduate Researcher (Advisor: Dr. Anita Hopper)

Project: Free introns of tRNAs (fitRNAs) as novel noncoding regulatory RNAs

Scripps Research, San Diego, CA

May 2025 – August 2025

Research Intern (Advisors: Dr. Megan Ken & Dr. Jamie Williamson)

Project: Observing the assembly of HIV-1 Gag proteins on a supported lipid bilayer

National Cancer Institute, Frederick, MD

May 2024 – August 2024

Research Intern (Advisor: Dr. Sandra Wolin)

Project: Investigation of Y RNA-dependent Ro60 protein interactions in *Salmonella Typhimurium*

Research Fellowships and Awards

NIH T32 Training Grant Appointment

September 2026

National Science Foundation Graduate Research Fellowship – \$159,000

April 2026

– Proposal title: “Nanoscale architecture and dynamics of mammalian Pol III transcription factories”

Center for RNA Biology Trainee Spotlight

March 2026

Semi-Finalist: Hertz Foundation Graduate Fellowship

November 2025

RNA Society Annual Meeting Oral Presentation Prize

June 2025

Undergraduate Research Scholarship

April 2023, -24, -25

– Awarded three years consecutively totalling \$16,900

Invited Speaker at RNA Society Annual Meeting

April 2025

– Only undergraduate speaker; >1,000 attendees

Barry M. Goldwater Foundation Scholarship – \$7,500

March 2025

RNA Society Research Presentation Fellowship – \$1,000

February 2025

Invited Speaker at Rustbelt RNA Meeting Undergraduate Symposium

October 2024

Eleanor Ruffing McMahon Conference Travel Award

October 2024

NCI Cancer Research Training Award – \$8,900

May 2024

Center for RNA Biology Symposium Best Poster Award

May 2024

STEP Fellowship – \$2,000

May 2024

Pelotonia Foundation Undergraduate Cancer Research Fellowship – \$12,000

April 2024

– One of 13 awarded in 2024

Thomas J. Byers Memorial Research Scholarship	March 2024
– <i>Declined due to mutually exclusive funding</i>	
Rustbelt RNA Meeting Poster Excellence Award	October 2023
Department of Molecular Genetics Travel Funding	September 2023
Jim Hopper Memorial Undergraduate Research Award	April 2023
– <i>Declined due to mutually exclusive funding</i>	

Publications

1. Nostramo RT, **Sinopoli PL**, Bao A, Metcalf S, Peltier LM, Hopper AK. (2025). Free introns of tRNAs as novel complementarity-dependent regulators of gene expression. **Molecular Cell**. PMID: 39938518.
Featured article of the issue. Highlighted by: Cell Press, OSU News, Science Direct, and others.
2. Metcalf S, Nostramo RT, Bao A, Clark K, **Sinopoli PL**, Hopper AK (2026). Many paths to destruction: Family-specific turnover and stress-responses for tRNA introns. **PNAS**. PMID: 41961856. Preprint on BioRxiv.

Oral Presentations

30th Annual Meeting of the RNA Society , San Diego, CA	June 2025
Invited Speaker – “Free tRNA introns act as novel complementarity-dependent noncoding regulatory RNAs”	
<i>Won RNA Society Oral Presentation Prize (1 awarded)</i>	
Rustbelt RNA Meeting Undergraduate Symposium , Newark, OH	October 2024
Invited Speaker – “Free tRNA introns act as novel noncoding regulatory RNAs”	

Select Poster Presentations

Denman Undergraduate Research Forum , Columbus, OH	March 2026
Poster – “tRNA introns: Ancient RNAs that hunt genetic messages”	
Scripps Research Summer Poster Session , La Jolla, CA	August 2025
Poster – “Observing the assembly of HIV-1 Gag proteins on a supported lipid bilayer”	
NIH Poster Day , Bethesda, MD	August 2024
Poster – “Investigation of Y RNA-dependent Ro60 protein interactions in <i>Salmonella Typhimurium</i> ”	
The Center for RNA Biology Symposium , Columbus, OH	May 2024
Poster – “fitRNAs: A new class of regulatory small noncoding RNAs”	
<i>Won Best Poster Award (5 awarded from 64 posters)</i>	
Rustbelt RNA Meeting , East Lansing, MI	October 2023
Poster – “tRNA introns act as novel noncoding regulatory RNAs”	
<i>Won Poster Excellence Award (4 awarded from 169 posters)</i>	
Department of Molecular Genetics Open House , Columbus, OH	September 2023
Poster – “The biological significance of tRNA introns”	

Science Outreach, Service, and Teaching

Introduction to Life Sciences: Research Opportunities and Career Options	August 2024 – October 2025
Undergraduate Panelist	
Phenomenon Design Summit , La Jolla, CA	July 2025
Speaker – “Biophysics and Structural Biology: the Assembly of Biomolecules” (tailored to K-12 teachers)	
– <i>Topic selected by science teachers for adaptation to high-school lesson plans</i>	

– Helped develop lesson plan “Virus Gone Viral” that is taught in San Diego High Schools

DNA Workshops in Columbus Public Schools

January 2025 – May 2025

Team Leader – Molecular Biology & Gel Electrophoresis Workshops

Village Book Builders

September 2023 – May 2025

Virtual science tutor for high school students in Kadzkalowa, Malawi

– Helped fundraise to re-open the local library after closure due to financial obstacles

TEDxOhioStateUniversity

August 2023 – May 2024

Committee Member

Inaugural Schoenberg Public Lectureship, Columbus, OH

November 2023

Public Outreach Poster – “Unconventional mRNA silencing in baker’s yeast: junk to gold?”